

The Cost Function for Logistic Regression

The goal of a **cost function**, $J(\vec{w}, b)$, is to measure how well a set of parameters $(\vec{w}, b)$ performs on the training data. A lower cost means a better fit.

Why Not Use the Squared Error Cost?

1. The Problem of Non-Convexity:

- If we use the same squared error cost function from linear regression, $J(\vec{w}, b) = \frac{1}{2m} \sum_{i=1}^m (f_{\vec{w}, b}(\vec{x}^{(i)}) - y^{(i)})^2$, with the logistic regression model's sigmoid function, the resulting cost function shape is **non-convex**.
- A non-convex function has many "dips" or **local minima**.
- This is a problem because **gradient descent** is not guaranteed to find the best possible parameters (the **global minimum**) and can get stuck in a suboptimal local minimum.

2. **Conclusion:** A different cost function is needed for logistic regression to ensure it is convex and that gradient descent can work reliably.

Logistic Loss Function

To build a better cost function, we first define the error on a *single* training example. This is called the **Loss Function**, denoted as $L(f_{\vec{w}, b}(\vec{x}), y)$.

The **Logistic Loss Function** is defined as follows:

- If the true label $y = 1$:
$$L(f(\vec{x}), 1) = -\log(f(\vec{x}))$$
 - If the true label $y = 0$:
$$L(f(\vec{x}), 0) = -\log(1 - f(\vec{x}))$$
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Intuition Behind the Logistic Loss 🧠

This loss function is designed to heavily penalize predictions that are confidently wrong.

- **When the true label is 1 ($y = 1$):**
 - If the model predicts $f(\vec{x}) \rightarrow 1$ (a correct, confident prediction), the loss $-\log(1)$ approaches **0**.
 - If the model predicts $f(\vec{x}) \rightarrow 0$ (an incorrect, confident prediction), the loss $-\log(0)$ approaches **infinity**.
- **When the true label is 0 ($y = 0$):**

- If the model predicts $f(\vec{x}) \rightarrow 0$ (a correct, confident prediction), the loss $-\log(1 - 0)$ approaches **0**.
 - If the model predicts $f(\vec{x}) \rightarrow 1$ (an incorrect, confident prediction), the loss $-\log(1 - 1)$ approaches **infinity**.
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The Full Cost Function

The overall **cost function** $J(\vec{w}, b)$ is simply the average of the loss function across all m training examples:

$$J(\vec{w}, b) = \frac{1}{m} \sum_{i=1}^m L(f_{\vec{w}, b}(\vec{x}^{(i)}), y^{(i)})$$

- This new cost function is **convex**, which guarantees that gradient descent can converge to the global minimum.